Discover Education

https://doi.org/10.1007/s44217-026-01577-9

Article in Press

Nocturnal internet use and its association with behavioral, psychological, and sociodemographic variables among university students in Bangladesh

Received: 27 November 2025

Accepted: 28 April 2026

Published online: 21 May 2026

Cite this article as: Khatun M.T., Mondal M.N.I., Mondal M.A.H. *et al.* Nocturnal internet use and its association with behavioral, psychological, and sociodemographic variables among university students in Bangladesh. *Discov Educ* (2026). https://doi.org/10.1007/s44217-026-01577-9

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Title: Nocturnal Internet Use and Its Association with Behavioral, Psychological, and Sociodemographic Variables among University Students in Bangladesh

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Running title: Nocturnal Internet Use

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Abstract

Background: The rise in nocturnal internet use (NIU) among university students has raised concerns about their mental health, academic performance, and sleep quality. Understanding the behavioral and socio-demographic factors associated with this issue is essential for developing effective interventions. However, the study aimed to examine the association of NIU with behavioral, psychological, and sociodemographic variables among university students.

Method: This was a cross-sectional study focusing on undergraduate students. The outcome variable was NIU, measured by participants' self-reported responses. The independent variables were measured using a standardized questionnaire that collected data on demographics, lifestyle choices, familial ties, and internet usage trends. Descriptive statistics, chi-square tests, and binary logistic regression were used to analyze the data.

Result: This study included 361 undergraduate students (59.3% male, 40.7% female). The prevalence of NIU was 71.2%. There was a strong association between NIU and father's graduate-level education (AOR=3.58; 95% CI: 1.270-10.082), unmarried marital status (AOR=2.80; 95% CI: 1.343-5.819), being in the fourth year (AOR=2.21; 95% CI: 1.242-3.919), having a "good" relationship with parents (AOR=1.79; 95% CI: 1.000-3.205), using the internet without studying (AOR=3.24; 95% CI: 1.542-6.785), and internet addiction (AOR=3.75; 95% CI: 1.925-7.303). The results suggest that both socio-demographic and behavioral factors were associated with nocturnal internet use among university students.

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Conclusion: Behavior, family, and academic factors all influence university students' nocturnal internet use. Targeted strategies that promote family support and responsible internet habits are necessary to reduce the negative impacts.

Keywords: Nocturnal internet use; Problematic internet use, Socio-demographic factors, University Students**,** Bangladesh

Introduction

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In today's digital era, the rapid growth of internet technology is vital to education, communication, entertainment, and more. Internet technology is one of the most important technological developments worldwide, and has become an integral part of people's daily lives [1]. Nocturnal Internet Use (NIU) refers to engaging in internet-based activities (e.g., browsing, social media, video watching, online communication) during the late evening or night hours that overlap with typical sleep periods, potentially delaying bedtime or reducing total sleep time and thereby contributing to disrupted sleep and related health or behavioral outcomes [2]. A study found that evening-type students (individuals who feel more energetic in the evening and follow a late sleep schedule) consistently showed the highest levels of late-night digital media use compared with morning (individuals who feel more productive at the start of the day and choose to earlier bedtime and

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wake up time) and intermediate types (individuals have no particular choices being active early or late in the day) [3, 4]. However, prolonged internet use at night can lead to irregular sleep cycles, disrupt the body's circadian rhythm, reduce concentration, cause health issues, and decline academic performance [5, 6]. Because of sleep disturbance, NIU is also highly linked with the risk of internet addiction, anxiety, depression, and social isolation [1, 7, 8]. In Bangladesh, the government aims to create a fully digitized and developed economy by 2041, where people can take advantage of the internet [9]. Easier and free internet access in Bangladesh may increase the propensity of individuals in this country to develop addiction-related symptoms as a consequence of internet use [1]. A quantitative study, 448 youths aged 15 to 35 were surveyed through stratified random sampling, 25.89% of the respondents were found to be internet addicts [10]. Recent studies showed that there is a higher variation in Asia in the prevalence of internet addiction among young people and adolescents, ranging from 8.1 to 50.9% [11-13]. However, it doesn't affect all individuals equally; instead, they vary across socio-demographic groups.

The NIU varies across all individuals and is influenced by socio-demographic variables such as age, gender, education, income, and residence. Males are generally more prone to internet addiction than females, though female overuse is rising [14]. Most males frequently use gaming and other platforms to earn money or find entertainment, while females prefer social media and

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communication platforms, as well as educational assistance [15]. Urban residents show higher NIU than rural individuals, and students with lower grades are more likely to develop internet addiction [14]. The study has found that the lower the father's educational level, the higher the prevalence of internet addiction [14]. Moreover, lower parental education, higher family income, and non-intact family structures are associated with increased internet addiction [16]. University students are particularly vulnerable to developing NIU because they have comparatively easy access to the internet via campus libraries and free campus wireless in internet zones. Focusing on university students is essential because increased age and greater autonomy from parental control in university life allow greater access to digital media [17]. The internet often becomes a tool for achieving these goals, but excessive reliance on it can gradually lead to maladaptive and problematic use [18]. However, this study aimed to investigate the associations of nocturnal internet use with behavioral, psychological, and socio-demographic variables. There are several reasons for conducting this research. This study focuses on university students because they are a population that is highly engaged with digital technology for both academic and recreational purposes. University students often face academic pressure, social influences, and flexible schedules, which may make them more susceptible to irregular internet habits and late-night usage. Ultimately, understanding these associations could help develop evidence-based strategies and

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interventions to encourage responsible internet use and protect physical and mental health.

Data and Methodology

Study design and settings

This study used a cross-sectional survey to examine the association between nocturnal internet use and socio-demographic factors among 3rd and 4th-year undergraduate students at Rajshahi University, one of Bangladesh's largest public universities, in the Rajshahi division, during July and August 2025. The research specifically targeted third and fourth-year students from various disciplines. These groups were chosen because they represent a diverse mix of academic experience and are frequent internet users, especially at night.

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Sample size and Procedure

The following equation was used to estimate the required participants for this study: $n = \frac{Z^2P(1-p)}{d^2}$, where n = sample size; z = 1.96 (95% CI); p = prevalence estimate (27.1%) found from a Bangladeshi study [19]); and d = precision of the prevalence estimate (0.05). So, the 304 sample is needed for this study. However, data from 361 participants were collected via a structured questionnaire on campus during class lectures, ensuring sufficient participants for robust statistical analysis.

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To ensure the accuracy and relevance of the data collected, only students actively enrolled at Rajshahi University during the data collection period were considered eligible. This guaranteed that respondents were currently engaged in the academic environment and had a digital lifestyle pertinent to the study. The survey was conducted during regular classroom sessions with the instructor's prior permission. Students attending these classes were invited to participate, creating a controlled setting for data collection and minimizing sampling bias. Participation was completely voluntary. Respondents who fully completed the questionnaires were included in the final analysis, while incomplete responses were excluded.

Data Collection

A structured questionnaire was used to collect data. To ensure clarity and reliability, the questionnaire was pre-tested before the survey. Data collection was conducted through the direct administration of the survey. Students completed the questionnaire in the researcher's presence to minimize non-response bias. Ethical considerations were strictly maintained. Participants were informed of the purpose of the study and assured that their responses would be kept confidential.

Measuring instrument

Internet Addiction Test (IAT)

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Internet addiction was assessed using the Internet Addiction Test (IAT) developed by Young [20]. The IAT consists of 20 items, each rated on a 5-point Likert scale ranging from 1(rarely) to 5 (always), with total scores ranging from 20 to 100. The cutoff score for the presence of internet addiction was 50 or above [21].

Kessler Psychological Distress (K-6)

Furthermore, psychological distress was measured using the Kessler Psychological Distress Scale (K-6) [22]. This includes six items assessing symptoms of anxiety and depressive distress experienced during the past 30 days. Each item is rated on a 5-point scale (0 = none of the time to 4 = all of the time), yielding a total score ranging from 0 to 24. The cutoff score for the presence of psychological distress was 13 or above [23].

Outcome variable

The outcome variable for this study was nocturnal internet use (NIU). Previous studies suggested that more than two hours daily and more than four days a week were significantly associated with problematic internet use [24, 25]. However, the cutoff point for NIU in this study was more than 4 days a week and 2 hours of internet use at night.

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Data Analysis

SPSS 22.0 was used for the statistical analysis. Descriptive statistics were used to describe the data and summarize respondents' socio-demographic characteristics and nocturnal internet use patterns, the dependent variable. We also checked the multicollinearity problem using the variance inflation factor (VIF); there are no multicollinearity problems if $0 \leq VIF \leq 5$. Additionally, the chi-square test and logistic regression analysis were applied

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to investigate the association between nocturnal internet use and socio-demographic variables and to identify associated factors, respectively. For this study, $p < 0.05$ was considered statistically significant.

Results

The Characteristics of the total respondents ($n = 361$) are presented in Table 1. Among the participants, 42.9% were 3rd-year students, and 57.1% were 4th-year students. The largest age group is 21-23 (62.3%), followed by 24-26 (37.7%). Among them, 59.3% were males while 40.7% were females. In terms of BMI, 8.6% had underweight, 76.2% had normal BMI, 12.5% were overweight, and 2.8% were obese. In this sample, 15.0% were married, and the rest (85.0%) were unmarried. Most of the participants (57.9%) had a good relationship with their parents. 34.6% had a friendly relationship with their parents. 37.4% participants' fathers completed a graduation degree. 31.3% participants' mothers passed the secondary exam. 12.2% of participants' fathers and 4.2% of participants' mothers completed a Master's degree. A large number (33.2%) of fathers' occupations were business, and (82.8%) of mothers' occupations were housewife. Per month, 65.9% families' income was ≤30000, and 4.1% was >30000. Among the participants, 80.3% belonged to nuclear families, while 19.7% were from joint families. A substantial proportion of the students, 68.7%, grew up in a village residence, and only 31.3% grew in an urban area. 60.1% students lived in a hall, 24.9% in a mess, and 15.0% at home. 69.5% students studied ≤4 hours per day. Among all the

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participants, 11.6% were smokers. Table 1 shows that 85.0% used the internet for study purposes. A large percentage of students (71.2%) used the internet late at night.

Table 1 also shows the association between nocturnal internet use and selected socio-demographic factors. A significant association was observed between late-night internet use and year of study ($p < 0.001$), with 4th-year students (78.2%) more likely to engage in late-night internet activity than 3rd-year students (61.9%). Marital status was significantly associated with NIU ($p < 0.001$). Unmarried participants (75.2%) reported a higher prevalence of nocturnal internet use than married participants (48.1%). Family-related characteristics showed notable associations. Participants with a bad relationship with their parents were more likely to engage in late-night internet activity ($p<0.05$). Similarly, father's educational qualification was strongly associated ($p < 0.001$) with a high prevalence among students whose fathers were graduates (87.4%). Moreover, mothers' educational qualifications were significantly related to the outcome variable ($p < 0.001$). Students whose mothers were graduates (85.4%) or higher educated (80.0%) had a higher prevalence of late-night internet use. In addition, smoking status was significantly associated ($p< 0.05$). Both internet addiction and psychological distress showed statistically significant associations with the outcome, with a higher prevalence observed among participants in the present compared to the absent group ($p<0.001$).

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Table 1: Background Characteristics and associations of NIU and selected variables (n = 361)

Variables	Levels	Frequency, *n* (%)	Nocturnal Internet use		P-value
			Yes, n (%)	No, n (%)	
Year	3rd year	155 (42.9)	96 (61.9)	59 (38.1)	**0.001**
	4th year	206 (57.1)	161 (78.2)	45 (21.8)	
Age	21-23	225 (62.3)	159 (70.7)	66 (29.3)	0.437
	24-26	136 (37.7)	98 (72.1)	38 (27.9)	
Gender	Male	214 (59.3)	158 (73.8)	56 (26.2)	0.112
	Female	147 (40.7)	99 (67.3)	48 (32.7)	
Body mass index	Underweight	31 (8.6)	21 (67.7)	10 (32.3)	0.962
	Normal	275 (76.2)	196 (71.3)	79 (28.7)	
	Overweight	45 (12.5)	33 (73.3)	12 (26.7)	
	obese	10 (2.8)	7 (70.0)	3 (30.0)	
Marital Status	Married	54 (15.0)	26 (48.1)	28 (51.9)	**<0.001**
	Unmarried	307 (85.0)	231 (75.2)	76 (24.8)	
Relations with parents	Friendly	125 (34.6)	77 (61.6)	48 (38.4)	**0.004**
	Good	209 (57.9)	156 (74.6)	53 (25.4)	
	Not good	27 (7.5)	24 (88.9)	3 (11.1)	
Father's educational qualification	Primary	65 (18.0)	42 (64.6)	23 (35.4)	**<0.001**
	Secondary	52 (14.4)	31 (59.6)	21 (40.4)	
	Higher secondary	65 (18.0)	39 (60.0)	26 (40.0)	
	Graduate	135 (37.4)	118 (87.4)	17 (12.6)	
	Masters	44 (12.2)	27 (61.4)	17 (38.6)	

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Father's occupation	Farmer Gov. employee Non-govt. employee Business Others	74 (20.5) 44 (12.2) 82 (22.7) 120 (33.2) 41 (11.4)	51 (68.9) 27 (61.4) 56 (68.3) 95 (79.2) 28 (68.3)	23 (31.1) 17 (38.6) 26 (31.7) 25 (20.8) 13 (31.7)	0.166
Mother's educational qualification	Primary Secondary Higher secondary Graduate Masters	85 (23.5) 113 (31.3) 100 (27.7) 48 (13.3) 15 (4.2)	57 (67.1) 67 (59.3) 80 (80.0) 41 (85.4) 12 (80.0)	28 (32.9) 46 (40.7) 20 (20.0) 7 (14.6) 3 (20.0)	**0.001**
Mother's occupation	Housewife Employee Others	299 (82.8) 41 (11.4) 21 (5.8)	214 (71.6) 30 (73.2) 13 (61.9)	85 (28.4) 11 (26.8) 8 (38.1)	0.612
Family income per month	≤ 30000 BDT > 30000 BDT	238 (65.9) 123 (34.1)	171 (71.8) 86 (69.9)	67 (28.2) 37 (30.1)	0.395
Family type	Nuclear Joint	290 (80.3) 71 (19.7)	212 (73.1) 45 (63.4)	78 (26.9) 26 (36.6)	0.072
Place of origin	Urban Village	113 (31.3) 248 (68.7)	75 (66.4) 182 (73.4)	38 (33.6) 66 (26.6)	0.108
Current living location	Home Mess Hall	54 (15.0) 90 (24.9) 217 (60.1)	36 (66.7) 58 (64.4) 163 (75.1)	18 (33.3) 32 (35.6) 54 (24.9)	0.125
Every day's study hour	≤ 4 >4	251 (69.5) 110 (30.5)	172 (68.5) 85(77.3)	79 (31.5) 25 (22.7)	0.058
Smoking status	No Yes	319 (88.4) 42 (11.6)	234 (73.4)	85 (26.6)	**0.012**

			23 (54.8)	19 (45.2)	
Using the internet without study purpose	No Yes	54 (15.0) 307 (85.0)	30 (55.6) 227 (73.9)	24 (44.4) 80 (26.1)	**0.006**
Internet Addiction Test	Absent Present	182 (50.42) 179 (49.58)	155 (85.2) 102 (57)	27 (14.8) 77 (43)	**0.000**
Psychological distress	Absent Present	147 (40.72) 214 (59.28)	88 (59.9) 169 (79)	59 (40.1) 45 (21)	**0.000**
Nocturnal internet use (total)			257 (71.2)	104 (28.8)	

In Table 2, the item number and mean of that item were indicated by the I and M, respectively. Table 2 reveals that students regularly stayed online longer than intended (I_{18}, M=3.09), lost sleep due to late-night internet use (I_{11}, M=2.93), repeatedly failed to cut down on internet use (I_{14}, M=2.92), and experienced academic consequences due to excessive use (I_3, M=2.89). These tendencies highlight a pattern of persistent difficulty in regulating internet engagement. Items associated with emotional reliance on the internet, such as fearing that life would be boring without it (I_9, M=2.73), anticipating the next opportunity to go online (I_8, M=2.59), and using the internet to avoid disturbing thoughts (I_7, M=2.45), were endorsed at occasional to frequent levels. It suggests that emotional dependence plays a notable role in students' overall internet use behavior. Interpersonal and behavioral conflict indicators, such as others complaining about excessive use (I_2, M=2.25), becoming irritated when interrupted while online (I_{10}, M=2.54), or choosing the internet over social interaction (I_{16}, M=2.20), were occasionally too frequent. Lower-scoring items generally related to relationship formation online (I_1, M=1.87) and to preferring internet use over intimacy (I_{20}, M=2.07), suggesting these behaviors are less common in this sample. The reliability analysis demonstrated strong internal consistency for

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the IAT in this population, with a Cronbach's alpha of 0.831, indicating that the items consistently measure the construct of internet addiction.

Table 2 Descriptive Statistics for Internet Addiction Test (IAT)

Items	**Mean**	**Std. Deviation**	**Description**
I_1: How often do you form new relationships with fellow online users?	1.87	1.098	Occasionally
I_2: How often do others in your life complain to you about the amount of time you spend online?	2.25	1.239	Occasionally
I_3: How often do your grades or school work suffer because of the amount of time you spend online?	2.89	1.297	Frequently
I_4: How often do you check your email before something else that you need to do?	2.30	1.136	Occasionally
I_5: How often does your job performance or productivity suffer because of the Internet?	2.70	1.172	Frequently
I_6: How often do you become defensive or secretive when anyone asks you what you do online?	2.42	1.193	Occasionally
I_7: How often do you block out disturbing thoughts about your life with soothing thoughts of the Internet?	2.45	1.089	Occasionally
I_8: How often do you find yourself anticipating when you will go online again?	2.59	1.125	Frequently
I_9: How often do you fear that life without the Internet would be boring, empty, and joyless?	2.73	1.205	Frequently
I_{10}: How often do you snap, yell, or act annoyed if someone bothers you while you are online?	2.54	1.110	Frequently
I_{11}: How often do you lose sleep due to being online?	2.93	1.211	Frequently
I_{12}: How often do you feel preoccupied with the Internet when off-line, or fantasize about being online?	2.25	1.234	Occasionally
I_{13}: How often do you find yourself saying "just a few more minutes" when online?	2.88	1.118	Frequently
I_{14}: How often do you try to cut down the amount of time you spend online and fail?	2.92	1.166	Frequently
I_{15}: How often do you try to hide how long you've been online?	2.38	1.208	Occasionally
I_{16}: How often do you choose to spend more time online over going out with others?	2.20	1.218	Occasionally

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I_{17}: How often do you feel depressed, moody, or nervous when you are off-line, which goes away once you are back online?	2.31	1.142	Occasionally
I_{18}: How often do you find that you stay online longer than you intended?	3.09	1.247	Frequently
I_{19}: How often do you neglect household chores to spend more time online?	2.59	1.183	Frequently
I_{20}: How often do you prefer the excitement of the Internet to intimacy with your partner?	2.07	1.169	Occasionally
Reliability Statistics Cronbach's Alpha (α): 0.831; **Mean**: 2.517			

Note: I_1-I_{20} indicates the number of items of IAT.

Binary logistic regression analysis (Table 3) shows several significant predictors of the outcome. 4th-year students were more than twice as likely as 3rd-year students (AOR: 2.21, p=0.007, 95% CI: 1.242-3.919) to use the internet late at night. Unmarried students were more than twice as likely to use the internet late at night compared to married students (AOR: 2.80, p=0.006, 95% CI: 1.343-5.819). Parental relationship quality showed a positive association, with 'good' relationships reaching borderline significance (AOR: 1.79, p=0.050; 95% CI: 1.000-3.205*).* Among parental education variables, only father's graduate-level education was significant (AOR: 3.58, p= 0.016, 95% CI: 1.270-10.082). Using the internet without study purposes (AOR: 3.24, p = 0.002, 95% CI: 1.542-6.785) and internet addiction (AOR: 3.75, p<0.001, 95% CI: 1.925-7.303) were significantly associated with NIU.

Table 3. Binary logistic regression analysis of variables associated with nocturnal internet use

Variables	**β**	***p-values***	**AOR, 95% CI**
Year (ref. = 3rd year)			
4th year	0.791	**0.007**	**2.21** (1.242-3.919)
Marital Status (ref. = Married)			
Unmarried	1.028	**0.006**	**2.80** (1.343-5.819)

Relations with parents (ref. = Friendly)			
Good	0.582	**0.050**	1.79 (1.000-3.205)
Not Good	1.403	0.059	4.06 (0.950-17.397)
Father's educational qualification (ref. =Primary)			
Secondary	-0.018	0.970	0.98 (0.388-2.488)
Higher secondary	0.096	0.855	1.10 (0.392-3.087)
Graduate	1.275	**0.016**	**3.58** (1.270-10.082)
Masters	-0.422	0.484	0.66 (0.201-2.138)
Mother's educational qualification (ref. = Primary)			
Secondary	-0.599	0.165	0.55 (0.235-1.280)
Higher secondary	-0.009	0.986	0.99 (0.363-2.707)
Graduate	0.462	0.488	1.59 (0.431-5.846)
Masters	0.656	0.475	1.93 (0.319-11.648)
Smoking status (ref. = No)			
Yes	-0.783	0.080	0.46 (0.190-1.099)
Using the internet without study purpose (ref. = No)			
Yes	1.174	**0.002**	**3.24** (1.542-6.785)
Internet Addiction Test (ref.= Absent)			
Present	1.322	**<0.001**	**3.75** (1.925-7.303)
Psychological distress (ref. = Absent)			
Present	0.128	0.682	1.14 (0.616-2.099)
Hosmer-Lemeshow test		χ^2=6.898	
		P=0.548	
Nagelkerke R Square		0.380	

Note: 'AOR, Adjusted Odds Ratio', 'CI, Confidence Interval', bold AOR indicates statistical significance

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Discussion

This study investigated the association between nocturnal internet use and socio-demographic variables among university students, with 71.2% students reporting late-night internet use. This is consistent with previous studies showing that young adults use the internet excessively late at night for entertainment, academic pressure, and social media engagement [26]. In Bangladesh, university students used the internet for communication and chatting purposes (90.6%), watching videos (87.7%), using social media (84.7%), and educational or informational purposes (82.2%) [1]. A previous study conducted in Bangladesh suggested that dissatisfaction with life, not engaging in social activities associated with internet addiction [27]. So, students who do not engage in social activities and are not satisfied with their lives may compensate by using the internet as a source of pleasure. They spent more time on the internet late at night (after 10:00 pm) [3]. As students advance in their studies, they may experience more social and academic demands, which could explain why fourth-year students are more likely than third-year students to use the internet at night. A study found that senior students frequently face increased workloads, deadline pressure, and career-planning stress, all of which may lead them to use the internet for extended periods as a diversion or to manage their academic work at night [28, 29]. This supports data from systematic reviews demonstrating a favorable association between academic stress and problematic online behavior, as well as irregular sleep schedules, among college students [30]. Marital status also turned out to be a significant predictor; single students had over three times the likelihood of using the internet at night. Unmarried students may use the internet in the evenings to find amusement and social connections, seeking social support through virtual friendships, which might encourage them to use the internet late into the night [2]. The family environment showed a strong association with nighttime internet use, especially the quality of the students' interactions with their parents. The likelihood of

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students using the internet at night was substantially higher for those with "good" parental relations than for those with friendly relationships. In this study, relationships with parents were classified into three groups: good (emotional warmth, trust, and regular communication), friendly (comfortable sharing and mutual respect), and not good (less communication and unresolved issues). These three types of family relationships were important because, in Bangladesh, young adults continue to be influenced by these relationships. Although the majority of university students live away from home, the perceived quality of relationships with parents influences lifestyle patterns, including nocturnal internet use. This emphasizes the protective or risk factors for behavioral health, including family dynamics and emotional support networks. Numerous studies have demonstrated that young people who have limited familial ties may turn to compensatory online socialization, which raises their risk of problematic nocturnal use [31]. Significant relationships were found between the participants' NIU and their parents' educational backgrounds. Students were more likely to use the internet late at night if their parents had a greater education (graduate). One possibility is that households with higher levels of education might have easier access to internet resources and digital devices. Parental education was also associated with greater rates of problematic internet behavior and late-night internet use [32]. This is consistent with research showing that problematic internet use is strongly linked to strained family dynamics as young people turn to the internet for escape or assistance [33]. Additionally, late-night internet use was substantially associated with both non-academic internet use and longer daily internet use (≥2 hours). Evidence that excessive or entertainment-driven internet use frequently persists into late hours, resulting in sleep difficulties, academic burnout, and decreased productivity [21, 32, 34]. Overall, the findings show that rather than being influenced by broad demographic factors, students' late-night internet use is largely

determined by their academic obligations, family dynamics, parental backgrounds, and personal lifestyle preferences.

Strengths and Limitations

This study presents several notable strengths. First, it adds to the limited research in Bangladesh examining the relationship between nocturnal internet use and socio-demographic, behavioral (internet addiction), and psychological (psychological distress) variables among university students. Second, the use of a relatively large and diverse sample size (n= 361) improves the statistical power and accuracy of the results. Using a structured, validated questionnaire within a cultural context ensures dependable data by yielding precise, comparable measurements. Third, applying robust statistical analysis techniques, including descriptive analysis, enhances the validity of the findings and offers a more comprehensive understanding of the links between nocturnal internet use and study variables. Finally, the study adhered to strict ethical standards, such as voluntary participation and confidentiality, thereby enhancing the credibility and reliability of the data collection process.

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Although this study has notable strengths, several limitations must be acknowledged. First, the cross-sectional design limits the ability to draw causal inferences about the relationship between nocturnal internet use and socio-demographic variables. The observed relationships only reflect a single point in time and cannot reveal long-term impacts. Second, the reliance on self-reported responses may introduce social desirability bias, as participants might underreport or overreport their internet habits. Third, since the study was conducted at only one university, its findings may not be generalizable to other institutions or regions of Bangladesh. Finally, some potential factors influencing NIU, such as loneliness, self-esteem, coping styles, screen light exposure, chronotype, and social isolation, were not examined in this study.

Practical Implications

The findings of this study have practical implications for educators, policymakers, and health professionals. This research can assist university authorities and counseling centers in designing awareness programs, digital literacy workshops, and sleep management interventions for students at the greatest risk. It emphasizes the need for campus-based mental health and wellness initiatives that address sleep disturbances, academic stress, and internet dependency. Moreover, students' advisors and faculty members can use these findings to monitor students' academic engagement and promote healthier online habits. From a policy standpoint, the results can help develop university guidelines for responsible internet use, encourage a balanced approach to technology, and enhance the overall academic environment.

Theoretical Implications

From a theoretical perspective, this study advances understanding of nocturnal internet use among young adults from behavioral and psychological standpoints. It aligns with frameworks such as the uses and gratifications theory, which suggests that individuals use the internet to fulfill specific needs, like social connection, entertainment, or academic assistance, but excessive use is linked to poor social relationships, lower life satisfaction, and decreased productivity [35, 36]. Additionally, the findings point toward future opportunities for theoretical development that connect internet behavior models to sleep and well-being theories, offering a more interdisciplinary view of the impacts of digital lifestyles.

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Conclusion

The study shows that university students often use the internet at night. This regular nighttime internet use is linked to behavioral and socio-demographic factors. Having a father with a graduate degree, being a senior, being single, and having a poor relationship with parents were all connected to NIU. Additionally, lifestyle factors like smoking status, daily online time, and the

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purpose of internet use had significant effects. These findings match earlier research indicating that family relationships, academic stress, and leisure internet use contribute to longer nighttime internet use, which can negatively affect students' sleep and overall well-being. To reduce the harmful effects of nighttime internet use, it is important to offer support to senior students, foster healthy family connections, and promote responsible internet habits, especially for leisure use. Future research should explore the psychological and cultural factors underlying this issue to develop tailored strategies for healthier internet use among university students.

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Human Ethics and Consent to Participate declarations: Written consent was taken from every participant before they participated in the study.

Clinical trial number: not applicable.

AI declaration

We used the **Grammarly premium version** for enhanced grammar, improving clarity, accuracy, conciseness, and tone.

Declarations

Ethical approval

Ethical approval for this study was obtained from the **Institutional Animal, Medical Ethics, Biosafety, and Biosecurity Committee** (IAMEBBC), Institute of Biological Sciences (IBSc), University of Rajshahi, Bangladesh (Memo No.: 131/320(49)/IAMEBBC/IBSc, dated May 26, 2025). All research procedures were conducted in accordance with the committee's ethical guidelines and regulations, and all methods were performed in compliance

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with institutional requirements and recognized standards for research involving human participants.

Consent to participate: All respondents provided informed written consent, acknowledging the study purpose and their rights, including confidentiality, anonymity, and the option to withdraw without explanation.

Consent to publish: The manuscript does not include identifiable individual data requiring consent for publication.

Availability of Data and Materials

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Funding statement: This research received no specific grant from any funding agency in the public, commercial, or non-profit sectors.

Conflict of interest statement: The authors declare no conflict of interest.

Acknowledgement: The authors thank all participants for their cooperation.

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